# Arbitrarily Long Gaps in OEIS A333635

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**Abstract**

OEIS A333635 consists of positive integers $n$ for which $n^2+1$ has at most two prime factors, counted with multiplicity. Iwaniec's theorem establishes that this set is infinite. We give a complementary unconditional fact: for every $L \geq 1$, infinitely many blocks of $L$ consecutive positive integers contain no term of A333635. In fact, the initial points of these blocks can be chosen in any prescribed arithmetic progression. The proof uses primes $p \equiv 1 \pmod 4$ and the Chinese remainder theorem.

**Keywords:** almost-primes; quadratic polynomials; prime divisors; Chinese remainder theorem; OEIS A333635.

## 1. The sequence and a gap theorem

Let $\Omega(m)$ count prime factors of $m$ with multiplicity and define

$$\mathcal{S} = \{n \in \mathbb{Z}_{>0} : \Omega(n^2+1) \leq 2\}.$$

The increasing enumeration of $\mathcal{S}$ is OEIS A333635 [1]. The infinitude of $\mathcal{S}$ was established by Iwaniec's theorem on almost-primes represented by quadratic polynomials [2]. We prove a complementary spacing assertion.

**Theorem 1.** *Fix positive integers $L, d$ and an integer $b$. There are infinitely many positive integers $N \equiv b \pmod d$ such that*

$$\Omega((N+j)^2+1) \geq 3 \qquad (0 \leq j < L). \tag{1}$$

*In particular, $\mathcal{S}$ has arbitrarily large gaps between successive terms.*

## 2. Proof by simultaneous quadratic congruences

*Proof.* Choose $3L$ pairwise distinct primes

$$p_{j,t} \equiv 1 \pmod 4, \qquad 0 \leq j < L, \quad t \in \{1,2,3\},$$

none of which divides $d$. Such a choice is possible because Dirichlet's theorem gives infinitely many primes in the progression $1 \pmod 4$.

For each such prime $p_{j,t}$, the congruence

$$x^2 \equiv -1 \pmod{p_{j,t}}$$

has a solution $r_{j,t}$: indeed, $-1$ is a quadratic residue modulo every prime $p \equiv 1 \pmod 4$, by Euler's criterion. Impose the simultaneous congruences

$$N \equiv b \pmod d, \qquad N \equiv r_{j,t} - j \pmod{p_{j,t}} \quad (0 \leq j < L,\ 1 \leq t \leq 3). \tag{2}$$

Because the auxiliary primes are pairwise distinct and coprime to $d$, the Chinese remainder theorem provides a solution class modulo

$$M = d \prod_{j=0}^{L-1} \prod_{t=1}^{3} p_{j,t}.$$

Every positive representative $N$ of this class satisfies

$$(N+j)^2 + 1 \equiv r_{j,t}^2 + 1 \equiv 0 \pmod{p_{j,t}} \qquad (1 \le t \le 3).$$

For each $j$, the three distinct primes $p_{j,1}, p_{j,2}, p_{j,3}$ therefore divide $(N+j)^2 + 1$. Consequently (1) holds. Since the solution class has infinitely many positive representatives, these yield infinitely many such blocks.

Each integer $N, N+1, \ldots, N+L-1$ lies outside $\mathcal{S}$. As $\mathcal{S}$ itself is infinite by Iwaniec's theorem, its successive members must have gaps of size at least $L+1$ for arbitrarily large $L$. This proves the final assertion. □

## 3. Scope, priority and AI disclosure

Iwaniec's 1978 theorem supplies infinitely many terms of A333635; Theorem 1 shows that those terms have unbounded gaps. Neither statement proves that $n^2+1$ is prime infinitely often, which remains open. The construction is an application of classical congruence methods, with no claim of novelty or priority. The argument was generated and checked internally using AI; it has *not* undergone independent refereeing or formal certification.

## References

[1] The OEIS Foundation Inc., *A333635: Numbers m such that $m^2+1$ has at most 2 prime factors*, `https://oeis.org/A333635`, consulted 11 November 2026.

[2] H. Iwaniec, "Almost-primes represented by quadratic polynomials," *Inventiones Mathematicae* **47** (1978), no. 2, 171–188. `https://doi.org/10.1007/BF01578070`.

[3] The Chinese remainder theorem; Dirichlet's theorem on primes in arithmetic progressions; Euler's criterion (classical results).